Project Meeting | March 23, 2026

Agenda

1. 3DGS
2. 3DOD in snow
3. 3DOD in night&rain

1. 3DGS Progress Update

- LongSplat Test With WeatherGS Dataset

2. 3DOD in snow Progress Update

- LiDAR model eval
  - PointPillars

Repo	Metric	Pedestrian	Cyclist	Car	GPU
PointPillars_repo	3D-Bbox	51.4642 47.9446 43.8040	81.8677 63.6617 60.9126	86.6456 76.7439 74.1668	1080ti x1
PointPillars_kitti	3D-Bbox	51.4551 47.9575 43.8407	81.8821 63.6698 60.9022	86.6348 76.7492 74.1609	5060ti x1
PointPillars_kitti_snow	3D-Bbox	50.9077 45.9942 42.0627	70.2602 44.4837 42.9534	55.0510 37.5991 32.3090	5060ti x1
PointPillars_repo	BEV	59.1687 54.3456 50.5023	84.4268 67.1409 63.7409	89.9664 87.9145 85.7664	
PointPillars_kitti	BEV	59.1745 54.3432 50.5029	84.4268 67.1347 63.7455	89.9664 87.9104 85.7638	
PointPillars_kitti_snow	BEV	56.2570 50.6632 47.2526	71.4513 46.5849 44.2080	74.9016 50.8983 49.5825	
PointPillars_repo	2D-BBox	64.6249 61.4201 57.5965	86.2569 73.0828 70.1726	90.6471 89.3330 86.6583	
PointPillars_kitti	2D-BBox	64.6159 61.3713 57.6031	86.2569 73.0707 70.1706	90.6471 89.3323 86.6528	
PointPillars_kitti_snow	2D-BBox	56.9435 51.8297 49.3069	76.5018 52.2800 49.3459	78.9322 59.3901 51.9992	
PointPillars_repo	AOS	49.3777 46.7284 43.8352	85.0412 69.1024 66.2801	90.4752 88.6828 85.7298	
PointPillars_kitti	AOS	49.2289 46.5926 43.7520	85.0412 69.0913 66.2914	90.4754 88.6834 85.7248	
PointPillars_kitti_snow	AOS	41.4645 38.1470 36.4075	75.5731 50.5951 47.5046	77.9229 58.2583 50.6567	

2. 3DOD in snow Progress Update

- LiDAR model eval
  - PointRCNN

Repo	Metric	Pedestrian AP@0.5	Cyclist AP@0.5	Car AP@0.7	GPU
PointRCNN_repo	3D-Bbox	-	-	-	-
PointRCNN_kitti	3D-Bbox	70.3937, 62.4997, 56.4812	86.8038, 67.5356, 64.1532	84.0957, 75.6573, 75.8359	5060ti x1
PointRCNN_kitti_snow	3D-Bbox	69.2039, 61.6143, 55.8719	79.7640, 56.6199, 53.0094	64.7429, 52.0929, 48.6623	5060ti x1
PointRCNN_repo	BEV	-	-	-	
PointRCNN_kitti	BEV	73.7393, 65.3820, 58.8687	87.0454, 69.4767, 65.7828	87.8752, 85.8902, 85.3261	
PointRCNN_kitti_snow	BEV	73.4029, 65.4532, 59.0363	83.5402, 58.0971, 54.9253	77.3060, 64.4550, 59.9026	
PointRCNN_repo	2D-BBox	-	-	-	
PointRCNN_kitti	2D-BBox	71.4616, 65.1817, 59.4463	89.1654, 72.4245, 68.6297	94.9392, 87.9672, 88.0076	
PointRCNN_kitti_snow	2D-BBox	72.7548, 65.3306, 59.3042	83.7134, 58.8943, 56.7163	85.9689, 68.0673, 66.3882	
PointRCNN_repo	AOS	-	-	-	
PointRCNN_kitti	AOS	69.10, 62.54, 56.73	89.09, 71.76, 68.01	94.89, 87.81, 87.75	
PointRCNN_kitti_snow	AOS	69.94, 62.63, 56.60	83.64, 58.65, 56.48	85.33, 67.30, 65.22	

2. 3DOD in snow Progress Update

- LiDAR model eval
  - VoxelRCNN

Repo	Metric	Pedestrian AP@0.5	Cyclist AP@0.5	Car AP@0.7	GPU
VoxelRCNN_repo	3D-Bbox	-	-	-	-
VoxelRCNN_kitti	3D-Bbox	66.5348, 60.2782, 55.6542	85.9466, 72.1112, 68.8626	89.6149, 83.8685, 78.8079	5060ti x1
VoxelRCNN_snow	3D-Bbox	66.6909, 59.3986, 55.9992	83.5091, 59.1099, 55.1060	74.8540, 56.2723, 50.7544	5060ti x1
VoxelRCNN_repo	BEV	-	-	-	
VoxelRCNN_kitti	BEV	67.7843, 63.0610, 59.7228	92.3742, 74.5277, 72.2761	90.2165, 88.1085, 87.6400	
VoxelRCNN_snow	BEV	70.8167, 64.0834, 58.8447	87.0567, 60.0769, 57.7704	79.2096, 66.7585, 60.1988	
VoxelRCNN_repo	2D-BBox	-	-	-	
VoxelRCNN_kitti	2D-BBox	75.4673, 71.4710, 67.2322	94.7069, 82.7996, 80.4283	97.5992, 89.6159, 89.2497	
VoxelRCNN_snow	2D-BBox	75.7812, 68.8897, 66.4404	88.2552, 64.6207, 60.7499	89.7385, 71.0313, 70.0774	
VoxelRCNN_repo	AOS	-	-	-	
VoxelRCNN_kitti	AOS	70.46, 66.24, 62.10	94.51, 82.32, 79.90	97.55, 89.50, 89.06	
VoxelRCNN_snow	AOS	70.80, 64.04, 61.58	87.88, 64.09, 60.24	89.37, 70.42, 69.21	

2. 3DOD in snow Progress Update

- Multimodal model eval
  - mvmm

Repo	Metric	Pedestrian AP@0.5	Cyclist AP@0.5	Car AP@0.7	GPU
MVMM_repo	3D-Bbox	-	-	-	-
MVMM_kitti	3D-Bbox	58.0598, 54.6385, 51.5646	76.9789, 63.7341, 61.3158	88.7584, 78.1791, 77.2052	5060ti x1
MVMM_kitti_snow	3D-Bbox	53.9292, 52.8448, 50.2185	71.5915, 50.4884, 47.6168	74.8244, 54.8129, 50.4703	5060ti x1
MVMM_repo	BEV	-	-	-	
MVMM_kitti	BEV	61.9839, 58.5879, 55.5531	82.6586, 69.0585, 65.9598	90.1502, 87.6711, 86.3628	
MVMM_kitti_snow	BEV	60.4902, 58.1564, 55.3361	78.2400, 55.0631, 52.3427	83.7933, 65.5323, 60.4207	
MVMM_repo	2D-BBox	-	-	-	
MVMM_kitti	2D-BBox	66.6428, 64.5338, 62.9940	89.8408, 77.7837, 75.5444	94.6267, 89.7352, 89.1435	
MVMM_kitti_snow	2D-BBox	66.9772, 64.8389, 62.6820	82.8715, 60.3086, 58.3068	87.8108, 69.2236, 67.2224	
MVMM_repo	AOS	-	-	-	
MVMM_kitti	AOS	61.9944, 59.8307, 57.8615	89.6691, 76.8684, 74.5149	94.5895, 89.5287, 88.8404	
MVMM_kitti_snow	AOS	62.7076, 60.2084, 57.7439	82.6868, 59.9001, 57.8186	87.5130, 68.5216, 66.3796	

3. 3DOD in night&rain Progress Update

- Literature Review Overview

- **MoME: 디코더를 따로 두고, 쿼리별로 가장 믿을 수 있는 디코더를 선택하는 방식**

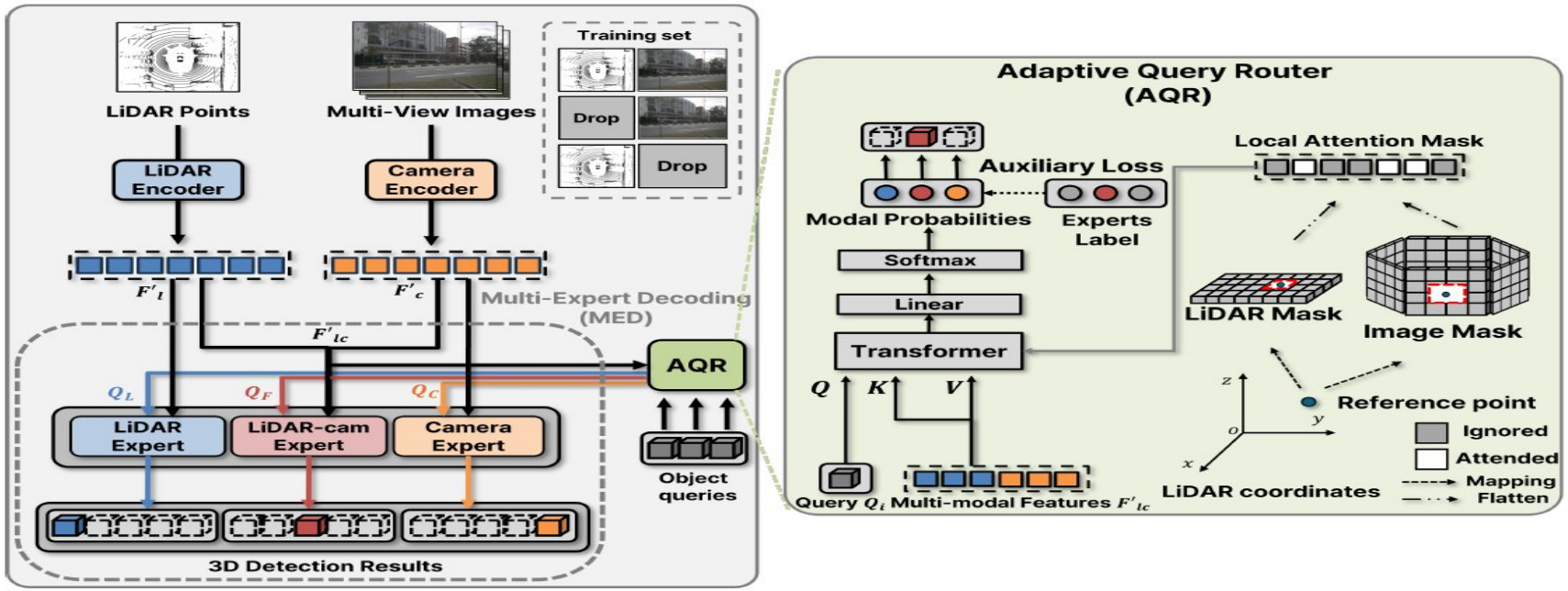


Figure 3. **Overall structure of MoME.** MoME utilizes three expert decoders, each specialized for processing LiDAR, camera, or LiDAR-camera features. The AQR dynamically assigns each object query to the most suitable expert decoder in an adaptive manner. The operation of AQR relies on local features filtered through the Local Attention Mask.

3. 3DOD in night&rain Progress Update

- Literature Review Overview

- **AWARDistill: 지식증류(KD) 기반 3D 탐지 프레임워크**

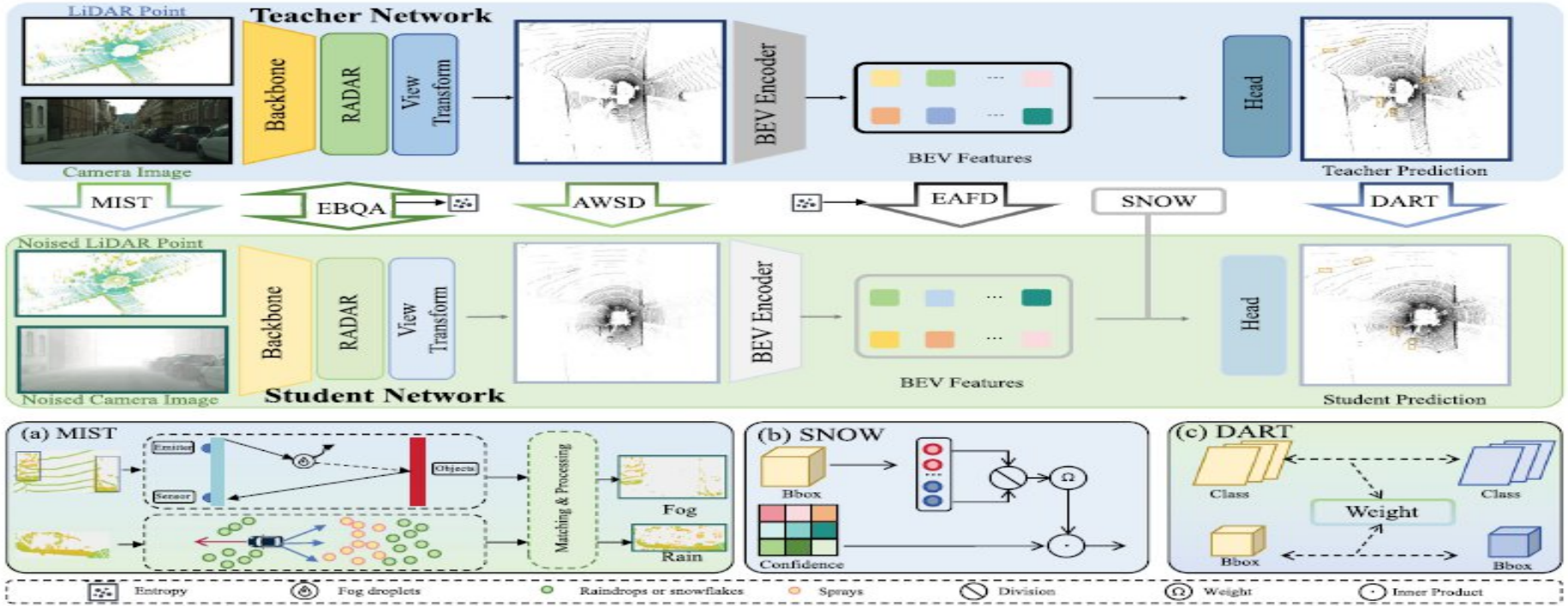


Fig. 3. The architecture of AWARDistill. (a), (b), and (c) denote the MIST, SNOW, and DART, respectively.

3. 3DOD in night&rain Progress Update

- Literature Review Overview

- **ContextualFusion: 낮/밤, 비/맑음 같은 환경 context를 이용해 LiDAR와 카메라의 비중을 동적으로 조절**하는 **gated fusion** 방식.

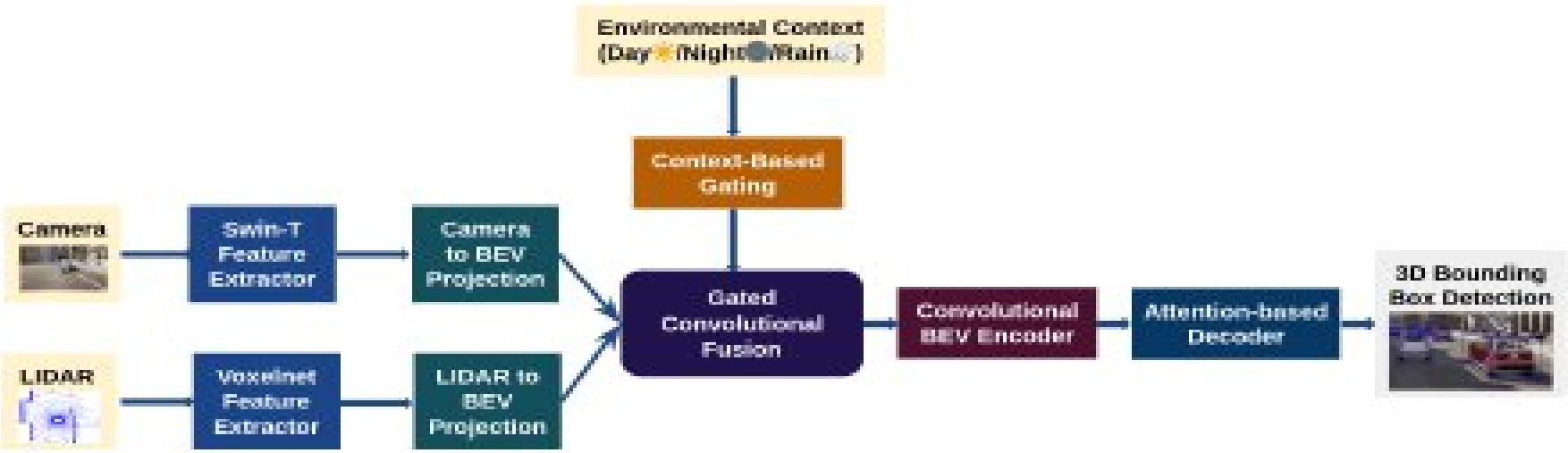


Fig. 4. Our *ContextualFusion* Model Architecture

3. 3DOD in night&rain Progress Update

- Literature Review Overview

- **Improving Robustness of LiDAR-Camera Fusion Model against Weather Corruption from Fusion Strategy Perspective**

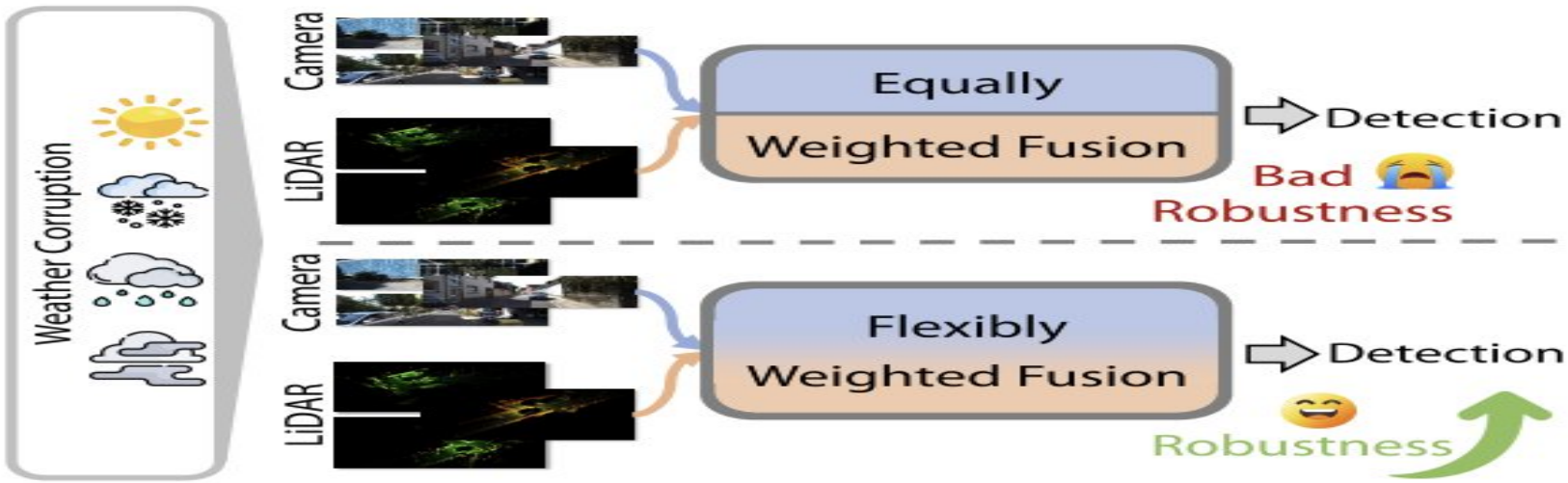
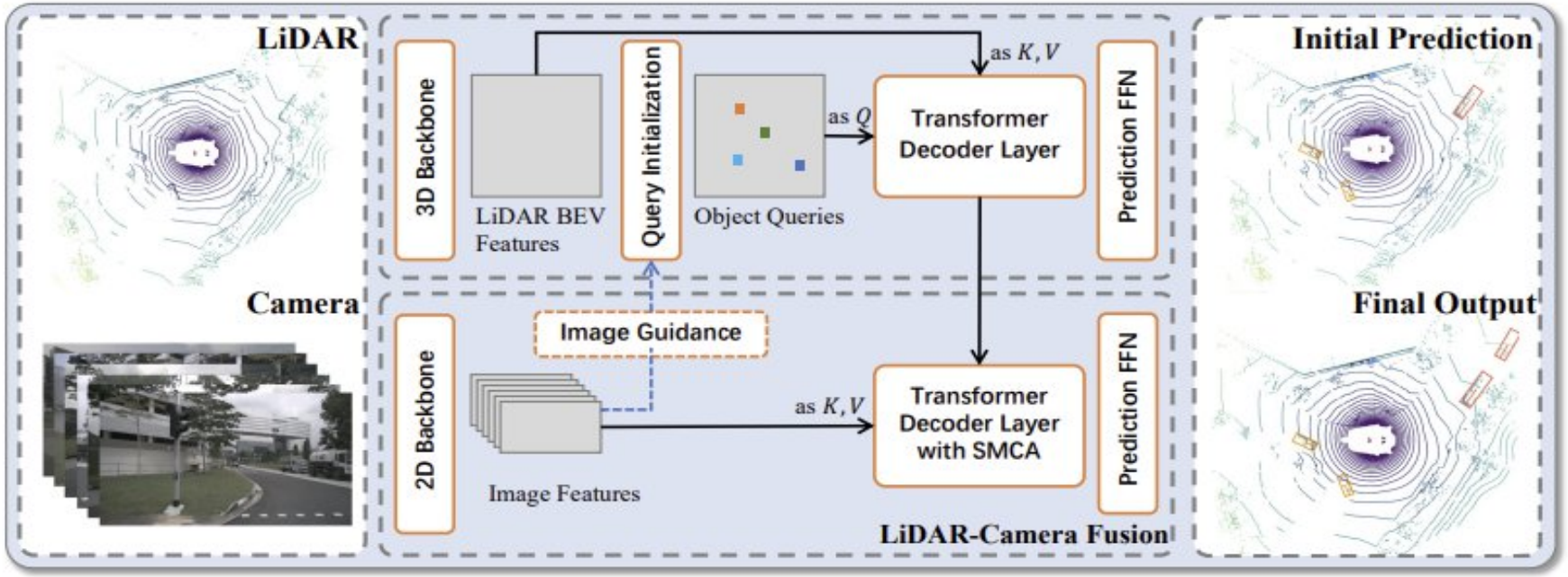


Fig. 1: Flexible weighted fusion between LiDAR and camera branches of the model improves the robustness of the LiDAR-camera fusion model against weather corruption.

3. 3DOD in night&rain Progress Update

- Literature Review Overview

- **TransFusion**: Transformer 기반으로 **object query와 이미지 /LiDAR feature를 soft하게 연결하는 fusion**



3. 3DOD in night&rain Progress Update

- Literature Review Overview

- 코드 및 weight 공개 여부

논문명	코드 공개	weight 공개	깃허브 링크
MoME	O	X	https://github.com/konyul/MoME
AWARDistill	X	X	X
ContextualFusion	O	O	https://github.com/ssuralcmu/ContextualFusion
Improving Robustness of LiDAR-Camera Fusion (FlexFusion)	X	X	X
TransFusion	O	X	https://github.com/XuyangBai/TransFusion